Multiple Linear Regression (Extra Examples)

CSU, Chico Math 314

2025-12-03

outline

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Data Set: PREVEND study

As adults age, cognitive function changes over time; largely due to various cerebrovascular and neurodegenerative changes.
The Prevention of REnal and Vascular END-stage Disease (PREVEND) study measured various clinical and demographic data for participants in a series of surveys between 1997 - 2006.

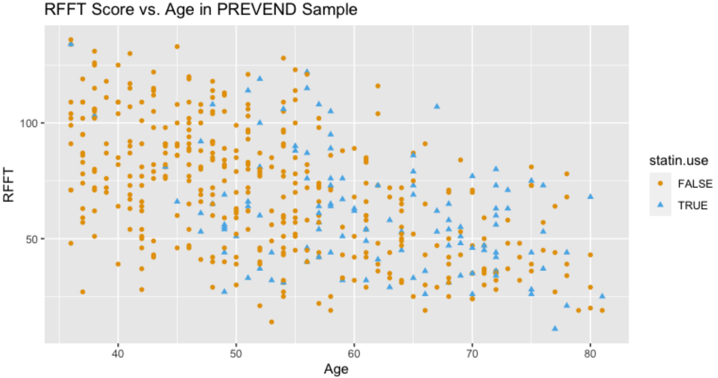
- Data from 4,095 participants are in the `prevend` dataset in the `oibiostat` package.
- Cognitive function was assessed with the Ruff Figural Fluency Test (RFFT), which provides information about cognitive abilities such as planning and the ability to switch between different tasks.
- Scores range from 0 to 175; higher scores indicate better cognitive function.
- We will work with a random sample of 500 participants.

Example Problem

Within the PREVEND data, we can explore the link between statin use and cognitive function

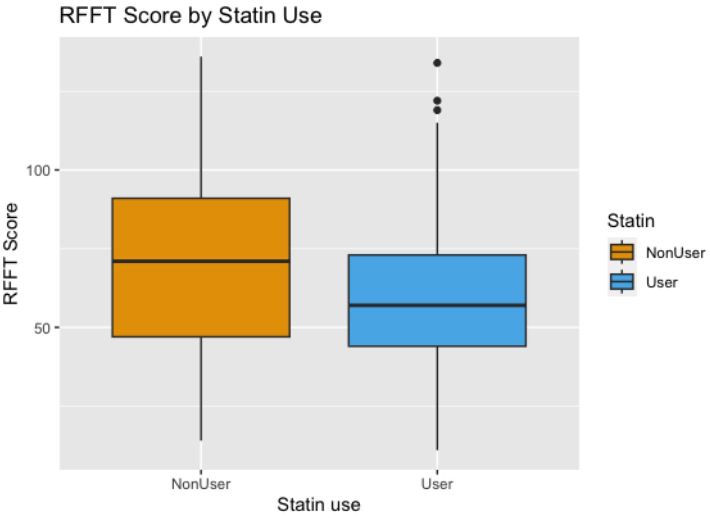
- Statins are a class of drugs widely used to lower cholesterol.
- If followed, recent guidelines for prescribing statins would lead to statin use in almost of Americans between 40 - 75 years of age and nearly all men over 60.
- A few small studies have suggested that statins may be associated with lower cognitive ability.
- The PREVEND study collected data on statin use as well as other demographic factors.

Age, Statin Use, and RFFT Score



- Blue triangles represent statin users; orange dots represent non-users.
- One research question was to examine the association between cognitive function and statin use after adjusting for age as a potential confounder.

RFFT Score vs. Statin Use



RFFT Score vs. Statin Use

```
#fit the linear model
lm(RFFT ~ Statin, data=prevend.samp)

##
## Call:
## lm(formula = RFFT ~ Statin, data = prevend.samp)
##
## Coefficients:
## (Intercept)   StatinUser
##       70.71       -10.05
```

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RFFT Score vs. Statin Use AND Age

```
#fit the linear model
lm(RFFT ~ Statin + Age, data=prevend.samp)

##
## Call:
## lm(formula = RFFT ~ Statin + Age, data = prevend.samp)
##
## Coefficients:
## (Intercept)   StatinUser          Age
##    137.8822       0.8509      -1.2710
```

RFFT Score vs. Statin Use AND Age

```
#print the model summary for the coefficients
summary(lm(RFFT ~ Statin + Age, data = prevend.samp))$coef

##                Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) 137.882219 5.1220851  26.91916 4.03116e-99
## StatinUser    0.850870 2.5957066   0.32780 7.43202e-01
## Age          -1.270995 0.0943036 -13.47769 1.65292e-35
```

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Confounding Variables (Reminder)

Similar to simple linear regression, we require the following assumptions:

- A confounding variable is a variable that is associated with both the explanatory and response variables.
- Example: Age would be considered a confounding variable when looking at the association between statin usage and RFFT score using the PREVEND dataset.

PREVEND: RFFT vs. Age (again)

We consider the hypothesis test from before, with $\alpha = 0.05$:

$$H_0 : \beta_1 = 0 \text{ (The X and Y variables ARE NOT associated)}$$

$$H_A : \beta_1 \neq 0 \text{ (The X and Y variables ARE associated)}$$

```
fit_slr <- lm(RFFT ~ Age, data = prevend.samp)
summary(fit_slr)$coef #p < 0.0001

##             Estimate Std. Err. t value Pr(>|t|)
## (Intercept) 137.5497 5.016141  27.421  1.4063e-101
## Age          -1.2614 0.089525 -14.089  3.5032e-38
```

With a p-value of $p < 0.0001$, we reject the null hypothesis and conclude that Age is linearly associated with RFFT.

PREVEND: Age vs. Statin Use

We consider a two-sample hypothesis test, as with the `fev` and `carnivora` data, with $\alpha = 0.05$:

$$H_0 : \mu_1 - \mu_2 = 0; H_1 : \mu_1 - \mu_2 \neq 0$$

This code will compare the mean age for the two levels of Statin use.

```
## Checking association between age and Statin
t.test(Age~Statin, data = prevend.samp)$p.value
#p < 0.0001

## [1] 8.716003e-14
```

With a p-value of $p < 0.0001$, we conclude that mean age for statin users and mean age for non-stain users are significantly different. This means we'll need to **adjust for age** when looking at the relationship between RFFT and Statin use.

Multiple Linear Regression in Action

We can use multiple linear regression to adjust for a confounding variable when the response variable is numerical.

```
fit_statinage <- lm(RFFT ~ Statin + Age,
data = prevend.samp)
summary(fit_statinage)

## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 137.8822     5.1221  26.919   <2e-16 ***
## StatinUser    0.8509     2.5957   0.328    0.743
## Age          -1.2710     0.0943 -13.478   <2e-16 ***

## Residual std err: 23.21 on 497 degrees of freedom
## Multiple R-squared:  0.2852, Adj. R-squared:  0.2823
## F-statistic: 99.13 on 2 and 497 DF,  p-value: < 2.2e-16
```

Multiple Linear Regression in Action (2)

We can use multiple linear regression to adjust for a confounding variable when the response variable is numerical.

```
fit_interaction <- lm(RFFT ~ Statin*Age,
data = prevend.samp)
summary(fit_interaction)
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     140.2031     5.6209  24.943   <2e-16 ***
## StatinUser      -13.9720    15.0113  -0.931    0.352
## Age              -1.3149     0.1040 -12.646   <2e-16 ***
## StatinUser:Age    0.2474     0.2468   1.003    0.317
## ---
## Residual std err: 23.21 on 496 degrees of freedom
## Multiple R-squared:  0.2866, Adj. R-squared:  0.2823
## F-statistic: 66.42 on 3 and 496 DF,  p-value: < 2.2e-16
```

Model Comparison with the stargazer Package

The stargazer() package is a useful way to display the summary of multiple models side-by-side. Here, we see how the models differ with and without an interaction term.

```
stargazer(fit_statinage, fit_interaction, ci = TRUE,
single.row = TRUE, type = "text", omit.stat = "ser",
title = "Results from multiple linear regression")
```

```
Results from multiple linear regression
=========================================================================
                                   Dependent variable:
               ----------------------------------------------------------
                                          RFFT
                           (1)                           (2)
-------------------------------------------------------------------------
StatinUser         0.851 (-4.237, 5.938)       -13.972 (-43.394, 15.450)
Age             -1.271*** (-1.456, -1.086)    -1.315*** (-1.519, -1.111)
StatinUser:Age                                   0.247 (-0.236, 0.731)
Constant       137.882*** (127.843, 147.921) 140.203*** (129.186, 151.220)
-------------------------------------------------------------------------
Observations               500                           500
R2                        0.285                         0.287
Adjusted R2               0.282                         0.282
F Statistic      99.132*** (df = 2; 497)       66.423*** (df = 3; 496)
=========================================================================
Note:                                        *p<0.1; **p<0.05; ***p<0.01
```

MLR - Unique Intercepts

The multiple regression model:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p$$

assumes that when one of the predictors x_j is changed by 1 unit and the values of the other variables remain constant, the predicted response changes by β_j, *regardless of the values of the other variables.*

For example, the equation for the multiple linear model predicting RFFT score from statin use and age based on the data in prevend.samp is:

$$\hat{RFFT} = 137.88 + 0.85 * StatinUser - 1.27 * Age$$

For Statin User ($StatinUser = 1$):

$$\hat{RFFT} = 137.88 + 0.85 - 1.27 * Age$$

For NonStatin User ($StatinUser = 0$):

$$\hat{RFFT} = 137.88 - 1.27 * Age$$

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Interaction (Definition)

A statistical interaction occurs when the effect of one explanatory variable x_j on the response variable y depends on the particular value(s) of one or more other explanatory variables.

In this section, we examine interaction in a two-variable case setting, where one of the predictors is categorical and the other is numerical.

Adding an Interaction Term

Consider the model:

$$E(RFFT) = \beta_0 + \beta_1(StatinUser) + \beta_2(Age) + \beta_3(StatinUser \times Age)$$

The term $(StatinUser \times Age)$ is the interaction term between statin usage and age, and β_3 is the coefficient of the interaction term.

Adding an Interaction Term in R

In R, we can use * or : to include an interaction term:

```
lm.interact <- lm(RFFT ~ Statin*Age, data = prevend.samp)
summary(lm.interact)$coef
##                  Estimate Std. Err.   t value   Pr(>|t|)
## (Intercept)      140.2031  5.62091  24.94312 1.3525e-89
## StatinUser       -13.9720 15.01132  -0.93077 3.5243e-01
## Age               -1.3149  0.10398 -12.64572 5.7438e-32
## StatinUser:Age     0.2474  0.24682   1.00255 3.1657e-01
```

Alternatively, the code below will produce the exact same results:

```
model.interact <- lm(RFFT ~ Statin + Age +
Statin:Age, data = prevend.samp)
summary(model.interact)$coef
```

Conclusion: The interaction term between statin usage and age is not statistically significant ($p = 0.32$). The relationship between age and RFFT does **not** depend on statin usage.

Another Example: Iris Dataset

The built-in `iris` dataset contains variables such as Petal.Length, Sepal.Length, and Species (3 levels).

```
table(iris$Species)
##
##     setosa versicolor  virginica
##         50         50         50
class(iris$Species)
## [1] "factor"
```

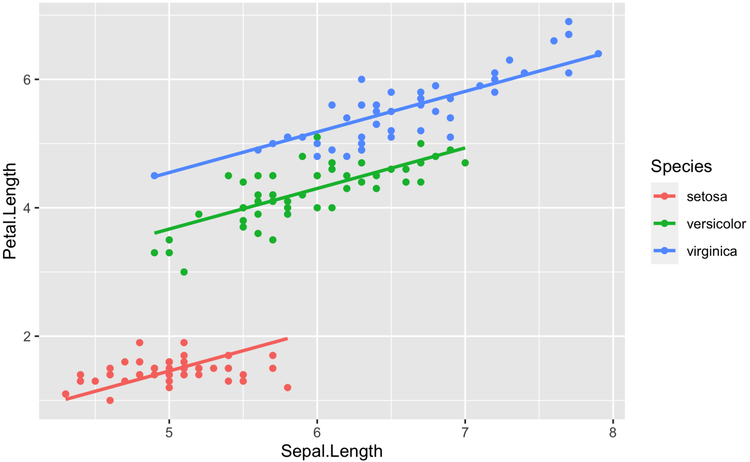
```
Call:
lm(formula = iris$Petal.Length ~ iris$Sepal.Length + iris$Species)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.76390 -0.17875  0.00716  0.17461  0.79954 

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)            -1.70234    0.23013  -7.397 1.01e-11 ***
iris$Sepal.Length       0.63211    0.04527  13.962  < 2e-16 ***
iris$Speciesversicolor  2.21014    0.07047  31.362  < 2e-16 ***
iris$Speciesvirginica   3.09000    0.09123  33.870  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2826 on 146 degrees of freedom
Multiple R-squared:  0.9749,    Adjusted R-squared:  0.9744 
F-statistic:  1890 on 3 and 146 DF,  p-value: < 2.2e-16
```

Data Visualization (no interaction term)



Unique intercepts, but the same slope.

Iris: Adding an Interaction Term with * or :

The built-in `iris` dataset contains variables such as Petal.Length, Sepal.Length, and Species (3 levels).

```
lm_iris2 <- lm(iris$Petal.Length ~
iris$Sepal.Length*iris$Species)

# Alternatively, we could use the code below to produce
the same result (including interaction term)

lm_iris2 <- lm(iris$Petal.Length ~
iris$Sepal.Length+iris$Species+
iris$Sepal.Length:iris$Species)
```

Iris: Adding an Interaction Term with * or :

The summary has separate slope terms for each Species:

```
summary(lm_iris2)
```

```
Call:
lm(formula = iris$Petal.Length ~ iris$Sepal.Length + iris$Species +
    iris$Sepal.Length:iris$Species)

Residuals:
     Min       1Q   Median       3Q      Max
-0.68611 -0.13442 -0.00856  0.15966  0.79607

Coefficients:
                                         Estimate Std. Error t value Pr(>|t|)
(Intercept)                                0.8031     0.5310   1.512    0.133
iris$Sepal.Length                          0.1316     0.1058   1.244    0.216
iris$Speciesversicolor                    -0.6179     0.6837  -0.904    0.368
iris$Speciesvirginica                     -0.1926     0.6578  -0.293    0.770
iris$Sepal.Length:iris$Speciesversicolor   0.5548     0.1281   4.330 2.78e-05 ***
iris$Sepal.Length:iris$Speciesvirginica    0.6184     0.1210   5.111 1.00e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2611 on 144 degrees of freedom
Multiple R-squared:  0.9789,    Adjusted R-squared:  0.9781
F-statistic:  1333 on 5 and 144 DF,  p-value: < 2.2e-16
```

Confirming Importance of Species with ANOVA

Using ANOVA, we confirm that petal length and sepal length are each different for at least one species in the iris data set.

```
summary(aov(iris$Petal.Length ~ iris$Species))
##               Df Sum Sq Mean Sq F value Pr(>F)
## iris$Species   2  437.1  218.55    1180 <2e-16 ***
## Residuals    147   27.2    0.19

summary(aov(iris$Sepal.Length ~ iris$Species))
##               Df Sum Sq Mean Sq F value Pr(>F)
## iris$Species   2  63.21  31.606   119.3 <2e-16 ***
## Residuals    147  38.96   0.265
```

Adding an Interaction Term

The complete model for all Species together can be expressed as follows:

$$\widehat{Petal.Length} =$$

$$0.80$$

$$+0.13(Sepal.Length)$$

$$-0.62(Speciesversicolor)$$

$$-0.19(Speciesvirginica)$$

$$+0.55(Sepal.length \times Speciesversicolor)$$

$$+0.62(Sepal.length \times Speciesvirginica)$$

What would the separate models be for each Species?

Significance of an Interaction Term

Taking the summary of the model with an interaction term, we find:

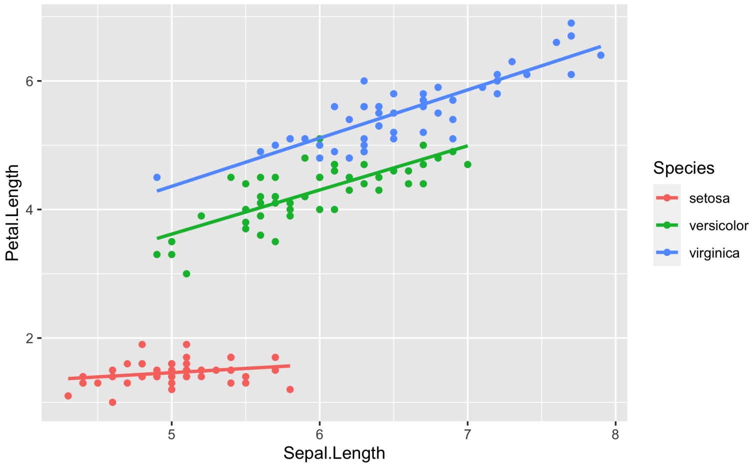
```
Call:
lm(formula = iris$Petal.Length ~ iris$Sepal.Length + iris$Species +
    iris$Sepal.Length:iris$Species)

Residuals:
     Min       1Q   Median       3Q      Max
-0.68611 -0.13442 -0.00856  0.15966  0.79607

Coefficients:
                                         Estimate Std. Error t value Pr(>|t|)
(Intercept)                                0.8031     0.5310   1.512    0.133
iris$Sepal.Length                          0.1316     0.1058   1.244    0.216
iris$Speciesversicolor                    -0.6179     0.6837  -0.904    0.368
iris$Speciesvirginica                     -0.1926     0.6578  -0.293    0.770
iris$Sepal.Length:iris$Speciesversicolor   0.5548     0.1281   4.330 2.78e-05 ***
iris$Sepal.Length:iris$Speciesvirginica    0.6184     0.1210   5.111 1.00e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2611 on 144 degrees of freedom
Multiple R-squared:  0.9789,    Adjusted R-squared:  0.9781
F-statistic:  1333 on 5 and 144 DF,  p-value: < 2.2e-16
```

Data Visualization (with interaction term)



Unique intercepts, unique slopes.

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Take Away

Multiple regression adds new layers of complexity to a relatively simple idea:

- fitting multiple lines across multiple explanatory variables,
- each line is interpreted with the other variables "held constant,"
- interaction terms are important to more accurately model real world problems
- Interpretation of adjusted R^2 remains
- checking model assumptions still necessary

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references I

David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel. *OpenIntro Statistics*. CreateSpace independent publishing platform, 4th edition, 2019.

Adapted from notes by Dr. Roualdes, Dr. Guo, and Dr. Charnigo